Status Report - CAREM (Argentina)

Published Date : Oct 16, 2024 10:02 AM CET

1. Overview

☐ **In Operation**

☑ **Under Construction**

☐ **Described in**

Link to PRIS	--

☐ **New Concept**

Number of Units	--
Year	--

Reference Plant

Name of Site	--
Link to PRIS	--
Net Power Output (MWe)	25
Gross Power Output (MWe)	29

1.1 Introduction

CAREM is a national SMR development project, based on LWR technology, coordinated by Argentina's National Atomic Energy Commission (CNEA) in collaboration with leading nuclear companies in Argentina with the purpose to develop, design and construct innovative small nuclear power plants with high level of safety and economic competitiveness. CAREM is an integral PWR type NPP, based on indirect steam cycle with features that simplify the design and support the objective of achieving a higher level of safety. CAREM25 is the demonstration prototype of CAREM SMR, and was developed using domestic technology, at least 70% of the components and related services for CAREM were sourced from Argentinean companies.

CAREM is designed as an energy source for electricity supply of regions with small demands. It can also support seawater desalination processes to supply water and energy to coastal sites.

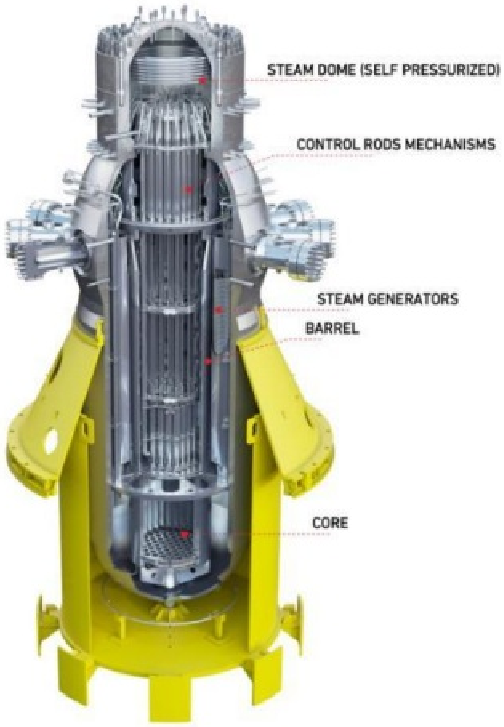

Figure 1. CAREM design

1.2 Development Milestones

1984	CAREM concept was presented in Lima, Peru, during the IAEA Conference on SMRs and was one of the first of the new generation reactor designs. CNEA officially launched the CAREM project
2006	Argentina Nuclear Reactivation Plan listed the CAREM25 project among priorities of national nuclear development
2009	CNEA submitted its preliminary safety analysis report (PSAR) for CAREM25 to the ARN
2011	Start-up of a high pressure and high temperature loop for testing the innovative hydraulic control rod drive mechanism
2011	Site excavation work beginning
2012	Civil design beginning
2013	Construction license
2014	Civil works started
2024	Electromechanical erection (start)
2028	First criticality

1.3 Design organization or vendor company

1.4 Links to designer/vendor website

1.5 Design Description

CAREM is a natural circulation based indirect-cycle reactor with features that simplify the design and improve safety performance. Its primary circuit is fully contained in the reactor vessel and it does not need any primary CAREM (CNEA, Argentina) All content included in this section has been provided by, and is reproduced with permission of CNEA, Argentina 22 recirculation pumps. The self-pressurization is achieved by balancing steam production and condensation in the vessel, without a separate pressurizer vessel. CAREM design reduces the number of sensitive components and potentially risky interactions with the environment. Some of the significant design characteristics are:

- Integrated primary cooling system; Self-pressurized; Core cooling by natural circulation; In-vessel control rod drive mechanisms; and Safety systems relying on passive features.

1.6 Licensing Application Support Documents

1.7 Reactor Unit in PRIS

2. Category Fields

Table 1: ARIS Category Fields

ARIS Category	Value	Units of Examples
Current/Intended Purpose	Commercial – Electric/Non-Electric	e.g. Commercial – Electric/Non-electric, Prototype/FOAK, Demonstration, Experimental
Main Intended Application (once commercial)	Baseload	e.g. Baseload, Dispatchable, Off-grid/Remote, Mobile/Propulsion, Non-electric (specify)
Reference Location	Inland	e.g. On Coast, Inland, Below-Ground, Floating-Fixed, Marine-Mobile, Submerged-Fixed (Other-specify)
Reference Site Design (reactor units per site)		e.g. Single Unit, Dual Unit, Multiple Unit (# units)
Core Coolant	H2O	H2O, D2O, He, CO2, Na, Pb, PbBi, Molten Salts, (Other-specify)

Reactor Core Size (1 core)	30 MWth <= Thermal power < 1000 MWth	e.g. Thermal Power < 30 MWth, 30 MWth ≤ Thermal power < 1000 MWth, 1000 MWth ≤ Thermal power ≤ 3000 MWth, Thermal power > 3000 MWth
Reactor Type	PWR	e.g. PWR, BWR, HWR, SCWR, GCR, GFR, SFR, LFR, MSR, ADS
Reactor Size	SMR	e.g. PWR, SMR
Neutron Moderator	H_2O	e.g. H_2O, D_2O, He, CO_2, Na, Pb, PbBi, Molten Salts, (Other-specify)
NSSS Layout	Integral	e.g. Loop-type (# loops), Direct-cycle, Semi-integral, Integral, Pool-type
Primary Circulation	Natural	e.g. Forced (# pumps), Natural
Thermodynamic Cycle	Rankine	e.g. Rankine, Brayton, Combined-Cycle (direct/indirect)
Secondary Side Fluid	H_2O	e.g. H_2O, He, CO_2, Na, Pb, PbBi, Molten Salts, (Other-specify)
Fuel Form	Fuel Assembly/Bundle	e.g. Fuel Assembly/Bundle, Coated Sphere, Plate, Prismatic, Contained Liquid, Liquid Fuel/Coolant
Fuel Lattice Shape	Hexagonal	e.g. Square, Hexagonal, Triangular, Cylindrical, Spherical, Other, n/a
Rods/Pins per Fuel Assembly/Bundle		e.g. #, n/a
Fuel Material Type	Oxide	e.g. Oxide, Nitride, Carbide, Metal, Molten Salt, (Other-specify)
Design Status	Under Construction	e.g. Conceptual, Detailed, Final (with secure suppliers)
Licensing Status	Under Construction	e.g. DCR, GDR, PSAR, FSAR, Design Licensed (in Country), Under Construction (# units), In Operation (# units)
Neutron Spectrum	Thermal	e.g. Fast, Thermal

3. Parameter Fields

Table 2: ARIS Parameter Fields

ARIS Parameter	Value	Units of Examples
Plant Infrastructure		
Design Life	40	Years
Lifetime Capacity Factor		%, defined as Lifetime MWe-yrs delivered / (MWe capacity * Design Life), incl. outages
Major Planned Outages		# days every # months (specify purpose, including refuelling)
Operation / Maintenance Human Resources		# Staff in Operation / Maintenance Crew during Normal Operation
Reference Site Design		n Units/Modules
Capacity to Electric Grid	25	MWe (net to grid)
Non-electric Capacity		e.g. MW(th) heat at x °C, m3/day desalinated water, kg/day hydrogen, etc.
In-House Plant Consumption		MWe
Plant Footprint	36000	m2 (rectangular building envelope)
Site Footprint		m2 (fenced area)
Emergency Planning Zone		km (radius)
Releases during Normal Operation		TBq/yr (Noble Gases / Tritium Gas / Liquids)
Load Following Range and Speed		x – 100%, % per minute
Seismic Design (SSE)	0.25	g (Safe-Shutdown Earthquake)
NSSS Operating Pressure (primary side)	12.25	MPa(abs), i.e. MPa(g)+0.1, at core outlet
NSSS Operating Pressure (secondary side)	4.7	MPa(abs), i.e. MPa(g)+0.1, at secondary side outlet
Primary Coolant Inventory (incl. pressurizer)		kg
Nominal Coolant Flow Rate (primary)		kg/s

Nominal Coolant Flow Rate (secondary)		kg/s
Core Inlet Coolant Temperature	284	ºC
Core Outlet Coolant Temperature	326	ºC
Available Tempeature as Process Heat Source		ºC
NSSS Largest Component		e.g. RPV (empty), SG, Core Module (empty/fuelled), etc.
- NSSS Largest Component Length		mm
- NSSS Largest Component Diameter		mm
- NSSS Largest Component Thickness		mm
- NSSS Largest Component Weight		tonne
Reactor Vessel Material		e.g. SS304, SS316, SA508, 800H, Hastelloy N
Steam Generator Design		e.g. Vertical/Horizontal, U-Tube/ Straight/Helical, cross/counter flow
Secondary Coolant Inventory		kg
Pressurizer Design		e.g. separate vessel, integral, steam or gas pressurised, etc.
Pressurizer Volume		m3 / m3 (total / liquid)
Containment Type		Dry (single/double), Dry/Wet Well, Inerted, etc.
Containment Volume		m3
Spent Fuel Pool Capacity and Total Volume		Years of full-power operation / m3
Non Electric Application		-
Core Delta Temperature	42	ºC
Fuel/Core		
Single Core Thermal Power	100	MW(th)
Refuelling Cycle	14	months or “continuous”

Fuel Material	UO2	e.g. UO2, MOX, UF4, UCO
Enrichment (avg./max.)	3.1	%
Average Neutron Energy		eV
Fuel Cladding Material		e.g. Zr-4, SS, TRISO, E-110, none
Number of Fuel “Units”		Specify as Assembly, Bundle, Plate, Sphere, or n/a
Weight of one Fuel Unit		kg
Total Fissile Loading (initial)		kg fissile material (specify isotopic and chemical composition)
% of fuel outside core during normal operation		Applicable to online refuelling and molten salt reactors
Fraction of fresh-fuel fissile material used up at discharge		%
Core Discharge Burnup	24	MWd/kgHM (heavy metal, eg U, Pu, Th)
Pin Burnup (max.)		MWd/kgHM
Breeding/Conversion Ratio		Fraction of fissile material bred in-situ over one fuel cycle or at equilibrium core
Reprocessing		e.g. None, Batch, Continuous (FP polishing/actinide removal), etc.
Main Reactivity Control	Control rods	e.g. Rods, Boron Solution, Fuel Load, Temperature, Flow Rate, Reflectors
Solid Burnable Absorber		e.g. Gd2O3
Core Volume (active)		m3 (used to calculate power density)
Fast Neutron Flux at Core Pressure Boundary		n/(cm²*s)
Max. Fast Neutron Flux		n/(cm²*s)
Core Control Rod Absorber		-
Core Soluble Neutron Absorber		-
Safety Systems		

Number of Safety Trains	Active / Passive	% capacity of each train to fulfil safety function
- reactor shutdown	Passive	- / - / -
- core injection	Passive	- / - / -
- decay heat removal	Passive	- /- / -
- containment isolation and cooling	Active/Passive	- / - / -
- emergency AC supply (e.g. diesels)	Active/Passive	- / - / -
DC Power Capacity (e.g. batteries)		Hours
Events in which Immediate Operator Action is required		e.g. any internal/external initiating events, none
Limiting (shortest) Subsequent Operator Action Time		Hours (that are assumed when following EOPs)
Severe Accident Core Provisions		e.g. no core melt, IVMR, Core Catcher, Core Dump Tank, MCCI
Core Damage Frequency (CDF)		x/reactor-year (based on reference site and location)
Severe Accident Containment Provisions		e.g., H2 ignitors, PARs, filtered venting etc.
Large Release Frequency (LRF)		x/reactor-year (based on reference site and location)
Overall Build Project Costs Estimate or Range (excluding Licensing, based on the Reference Design Site and Location)		
Construction Time (nth of a kind)		Months from first concrete to criticality
Design, Project Mgmt. and Procurement Effort		Person-years (PY) [DP&P]
Construction and Commissioning Effort		PY [C&C]
Material and Equipment Overnight Capital Cost		Million US$(2015) [M&E], if built in USA
Cost Breakdown	%[C&C]/%[M&E]	
- Site Development before first concrete		(e.g. 25/10) (30/40)
- Nuclear Island (NSSS)		

- Conventional Island (Turbine and Cooling)		(20/25) (20/10) (5/15) (-----------) (to add up to 100/100)
- Balance of Plant (BOP)		
- Commissioning and First Fuel Loading		
Factory / On-Site split in [C&C] effort		%/% of total [C&C] effort in PY (e.g., 60/40)
Nuclear System		
System Steam Flow Rate		kg/s
System Steam Pressure		MPa
System Steam Temperature		ºC
System Feedwater Flow Rate		kg/s
System Feedwater Temperature		ºC
Reactor Core		
Active Core Height		m
Equivalent Core Diameter		m
Average Linear Heat Rate		kW/m
Average Fuel Power Density		kW/kgU
Average Core Power Density		MW/m3
Outer Diameter Of Fuel Rods		mm

4. Plant Layout, Site Environment and Grid Integration

SUMMARY FOR BOOKLET

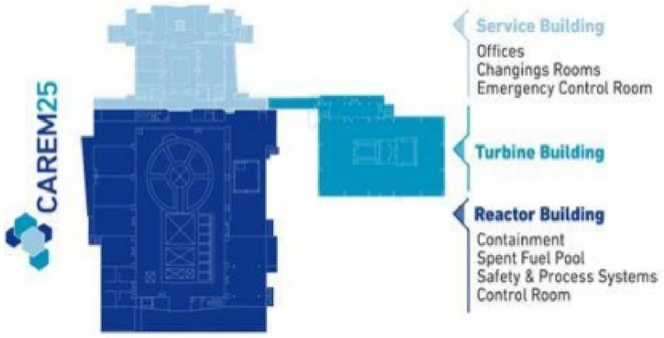

Figure 2. CAREM plant layout

Figure 3. CAREM25 construction site

4.1 Site Requirements during Construction
4.2 Site Considerations during Operation
4.3 Grid Integration

5. Technical NSSS/Power Conversion System Designn

SUMMARY FOR BOOKLET

Nuclear Steam Supply System

CAREM is an integral reactor. Its high-energy primary system (core, steam generators, primary coolant and steam dome) is within a single pressure vessel. Primary cooling flow is achieved by natural circulation, which is induced by placing the steam generators above the core. Water enters the core from the lower plenum. After being heated, the coolant exits the core and flows up through the chimney to the upper steam dome. In the upper part, water leaves the chimney through lateral windows to the external region. It then flows down through modular steam generators, decreasing its enthalpy.

Reactor Core

The reactor core of CAREM25 has hexagonal cross section fuel assemblies. There are 61 fuel assemblies with 1.4 meters active length. Each fuel assembly contains 108 fuel rods with 9 mm outer diameter, 18 guide thimbles and one instrumentation thimble. The fuel is 1.8% - 3.1% enriched UO_2. The fuel cycle can be tailored to customer requirements, with a reference design for the prototype of 390 full-power days and 50% of core replacement.

Reactivity Control

Core reactivity is controlled using Gd_2O_3 as burnable poison in specific fuel rods and movable absorbing elements belonging to the power adjustment and reactivity control system. Neutron poison in the coolant is not used for reactivity control during normal operation and in reactor shutdown. Each absorbing element consists of a cluster of rods linked to a structural element ('spider'), so the whole cluster moves as a single unit. Absorber rods fit into the guide tubes.

Reactor Pressure Vessel and Internals

Reactor Pressure Vessel (RPV) of CAREM25 is 11 meters high and 3.4 meters in diameter, with a variable thickness of 13cm to 20cm. The RPV is made of forged steel with an internal stainless steel liner.

Figure 4. RPV fabrication

Reactor Coolant System

The reactor´s coolant system of CAREM is fully contained in the RPV. Core cooling is achieved using natural circulation phenomenon with no primary coolant pumps. The reactor´s core heats the coolant causing it to flow upwards through the riser. Then it turns downwards to flow into the steam generators, reducing its temperature. Finally, it flows to the downcomer and the core, closing the circuit.

Steam Generator

In CAREM25, twelve (12) identical mini-helical vertical steam generators of the once-through type are placed equidistant from each other, along the inner surface of the RPV. Each consists of a system of 6 coiled piping layers, 52 parallel pipes of 26 m active length. They are used to transfer heat from the primary to the secondary circuit, producing superheated dry steam at 4.7MPa. The secondary system circulates upwards within the tubes, while the primary coolant moves in counter-current flow. The steam generators are designed to withstand the primary pressure. The entire secondary side is designed to withstand primary pressure up to isolation valves (including the steam outlet/water inlet headers) in case of SG tube breakage.

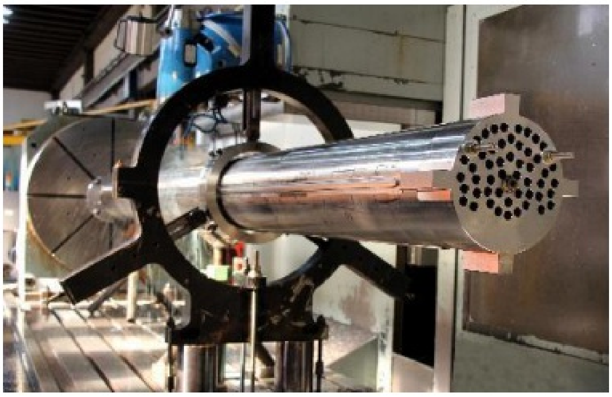

Figure 5. CAREM25 Steam generator fabrication

Pressurizer

Self-pressurization of the primary system in the steam dome is the result of the liquid–steam equilibrium. The large steam volume in the RPV, acting as an integral pressurizer, also contributes to damping of any pressure perturbations. Due to self-pressurization, the bulk temperature at core outlet corresponds to saturation temperature at primary pressure. In this way, typical heaters present in conventional PWR pressurizers are eliminated.

Instrumentation and Control Systems

Plant control is performed by a distributed control system, computer based and with high availability. There are two diverse protection systems: First Reactor Protection System (FRPS) and Second Reactor Protection System (SRPS), each system carries four redundancies. There are two (2) diverse nuclear instrumentation systems (NIS), one each for the FRPS and SRPS.

5.1 Primary Circuit
5.2 Reactor Core and Fuel
5.3 Fuel Handling
5.4 Reactor Protection
5.5 Secondary Side
5.6 Containment/Confinement
5.7 Electrical, I&C and Human Interface
5.8 Unique Technical Design Features (if any)

6. Technology Readiness

SUMMARY FOR BOOKLET

Testing Conducted for Design Verification and Validation

A full-scale loop has been built in order to test the innovative control rod drive mechanisms. This facility is operating at the same parameters (pressure and temperature) as RPV plant design conditions and is was designed for reactivity adjustment and control rods calibration.

Design and Licensing Status

CAREM25 basic design is completed. CAREM25 detail design is 90% progress. CAREM25 was granted a construction license. CNEA is working on the commissioning license request.

6.1 Deployed Reactors
6.2 Reactors under Licensing Review
6.3 Reactors in the Design Stage

7. Safety Concepts

SUMMARY FOR BOOKLET

Engineered Safety System Approach and Configuration

Defence in Depth (DiD) principle was the basis of this process. Appropriate criteria were defined for DiD internalization in the design taking into account CAREM general design characteristics, to prevent, control and mitigate the postulated events. Western European Nuclear Regulators Association proposal for DiD levels definition was adopted, which allows the consideration of Multiple Failures Events (Level 3B) as part of DiD Level 3 for the prevention of core damage. A strategy was defined for each one of the levels. For Level 2 the CVCS can control a medium loss of coolant event and LOHS. Two stages for level 3A and level 3B are established. For level 3A, the first stage is accomplished by means of passive safety systems, while the second stage is implemented by active systems in order to achieve the final safe state. For Level 3B, which is used in case of failure in Level 3A, the first stage is also accomplished by systems with passive processes while the second stage is composed by simple systems with external water supply. For Level 4 -postulated severe accident mitigation- provisions are considered to guarantee hydrogen control and RPV lower head cooling with the aim of in-vessel corium retention. CAREM´s safety system consists of two reactor protection systems (RPS), two shutdown systems, passive residual heat removal system (PRHRS), safety and depressurization valves, low pressure injection system and a containment of pressure suppression type. The First Shutdown System (FSS) consists of 9 fast shutdown rods and 16 reactivity adjust and control rods located over the core, which fall by gravity when needed. While the Second Shutdown System consists of a gravity assisted high-pressure injection of borated water from two high pressured tanks, which is activated automatically when failure of the FSS is detected. The different systems, structures and components (SSCs) that contribute to the reactor´s safety are classified according to the identification of low-level safety functions (LLSF) -derived from the fundamental safety functions- and safety functional groups of SSCs that fulfill those functions.

Decay Heat Removal System

During the grace period of 36 hours, core decay heat removal is assured by one out of two PRHRS in the case of loss of heat sink (LOHS) or Station Black-out (SBO). SBO is considered to be a design basis event. After that period, redundant active systems provide heat removal from the RPV and the containment to the final heat sink. Two redundant diesels provide energy supply for these systems. Despite the low frequency of an SBO longer than 36 hours, provisions for that scenario are supplied by simple systems supported by a fire extinguishing system or self-powered pumps. The PRHRS are heat exchangers formed by parallel horizontal U-tubes (condensers) coupled to common headers.

Emergency Core Cooling System

In case of a loss of coolant event, a Medium Pressure Injection System delivers water into the RPV to keep the core covered during the grace period. This system consists of two redundant borated water accumulators connected to the RPV. When the pressure in the reactor vessel becomes relatively low, the rupture disks, that isolate the accumulator tanks from the RPV will break. After the grace period and in case the loss of the coolant could not be isolated, an active system provides long term water injection into the RPV. Two redundant diesels provide energy supply for these systems.

Spent Fuel Cooling Safety Approach/System

In case of a SBO and during the grace period, longer than the reactor one, the water of the spent fuel pool is the heat sink. After this period, a chain of redundant active systems provides heat removal to the final heat sink. Two redundant diesels provide energy supply for these systems.

Containment System

The cylindrical containment vessel with a pressure suppression pool is a 1.2m thick reinforced concrete external wall with a carbon steel liner and withstands earthquakes of 0.25g. It is designed to withstand the pressure of 0.5MPa. The heat sink is located inside the containment, this provides protection for extreme external events during the grace period. After grace period, redundant active systems with diesel generator support, will provide suppression pool cooling and containment depressurization.

Plant Safety and Operational Performances

The natural circulation of coolant produces different flow rates in the primary system according to the power generated and removed. Under different power transients, a self-correcting response in the flow rate is obtained. 24 Due to the self-pressurizing of the RPV, the system keeps the pressure close to the saturation pressure. The control system is capable of keeping the reactor pressure practically at the operating set point through different transients, even in case of power ramps.

7.1 Safety Philosophy and Implementation
7.2 Transient/Accident Behaviour

8. Fuel and Fuel Cycle

SUMMARY FOR BOOKLET

Fuel Cycle Approach

The fuel cycle can be tailored to customer requirements, as a reference CAREM25 has an open fuel cycle design of 390 full-power days and 50% of core replacement.

Waste Management and Disposal Plan

Waste management facilities for waste treatment and storage are provided. Provisions for prolonged temporary storage of solid wastes at the site are considered.

8.1 Fuel Cycle Options
8.2 Resource Use Optimization
8.3 Unique Fuel/Fuel Cycle Design Features

9. Safeguard & Physical Security

SUMMARY FOR BOOKLET

9.1 Safeguards
9.2 Security
9.3 Unique Safeguards and/or Security Features

10. Project Delivery & Economics

SUMMARY FOR BOOKLET

As CAREM25 is a prototype, plant economics is not provided. CNEA is working on economic studies for the commercial module.

10.1 Project Preparation and Negotiation
10.2 Construction and Commissioning
10.3 Operation and Maintenance